随机过程

马啸

中山大学计算机学院

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Poission processes

Review

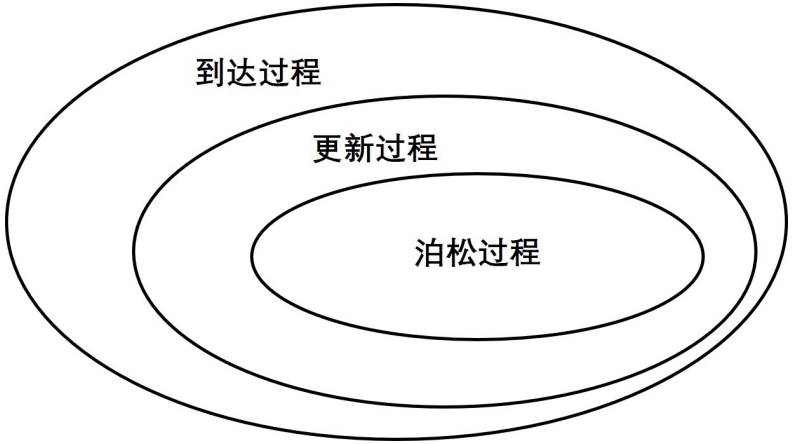
§ Combining and splitting Poisson processes

§ Non-homogeneous Poisson processes

§ Conditional arrival densities and order statistics

Review

- 到达过程：
  1. 随机变量序列 $0 < S_1 < S_2 < \ldots < S_n < \ldots$ (到达时刻)；
  2. $X_1, X_2, \ldots, X_n, \ldots$ (间隔过程), 其中 $X_1 = S_1$, $X_2 = S_2 - S_1, \ldots,$ $X_n = S_n - S_{n-1}$；
  3. $N(t), t \geq 0$, 表示 $(0, t]$ 时间区间内的到达数 (计数过程)。
- 更新过程：一种到达过程，满足：$X_1, X_2, \ldots, X_n, \ldots$ 是 i.i.d. 的。
- 泊松过程：一种更新过程，间隔变量满足：$P\{X > t\} = e^{-\lambda t}, t \geq 0$, 其中 λ 是到达速率。



Review

为刻画泊松过程，我们可以给出

$$S_1, S_2, \ldots, S_n \text{ 或 } X_1, X_2, \ldots, X_n \text{ 或 } N(t_1), N(t_2), \ldots, N(t_n)$$

的联合分布。我们的定义是从 X 出发的，而 X 是 i.i.d. 的，所以刻画一维分布即可。X 的分布函数为

$$F_X(t) \triangleq P\{X \leq t\} = 1 - P\{X > t\} = 1 - e^{-\lambda t},$$

密度函数为

$$f_X(t) = \lambda e^{-\lambda t},$$

其数学期望是 $\frac{1}{\lambda}$。

Review

$S_1, S_2, \ldots, S_n$ 的联合分布确定如下，其密度函数

$$f_{S_1,S_2,\ldots,S_n}(s_1, s_2, \ldots, s_n) = \lambda e^{-\lambda s_1} \cdot \lambda e^{-\lambda(s_2-s_1)} \ldots \lambda e^{-\lambda(s_n-s_{n-1})} = \lambda^n e^{-\lambda s_n},$$

$0 < s_1 < s_2 < \ldots < s_n$, 其边缘密度函数是 Erlang 分布

$$f_{S_n}(s) = \frac{\lambda^n s^{n-1} \exp(-\lambda s)}{(n-1)!}, s > 0.$$

$N(t)$ 表示 $(0, t]$ 中的到达数目，其服从泊松分布，分布律是

$$P_{N(t)}(n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, n = 0, 1, 2, \ldots$$

Review

泊松过程有三种定义：

Definition 1

到达时间间隔服从独立同分布的指数分布 $P\{X > t\} = e^{-\lambda t}, t \geq 0$。

Definition 2

计数过程是平稳独立增量过程，且 $(0, t]$ 之间的到达数目服从泊松分布 $P\{N(t) = n\} = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, n \geq 0$。

Definition 3

计数过程是平稳独立增量过程，且满足：

$$
\begin{aligned}
P\{\tilde{N}(t, t+\delta) = 0\} &\approx 1 - \lambda\delta + o(\delta) \\
P\{\tilde{N}(t, t+\delta) = 1\} &\approx \lambda\delta + o(\delta) \\
P\{\tilde{N}(t, t+\delta) = 2\} &\approx o(\delta)
\end{aligned}
$$

Poission processes

Review

§ **Combining and splitting Poisson processes**

§ Non-homogeneous Poisson processes

§ Conditional arrival densities and order statistics

Combining Independent Poisson Processes

Definition 4

Two Poisson processes $\{N_1(t) : t > 0\}$ and $\{N_2(t) : t > 0\}$ are independent if for all $t_1, \ldots, t_n$, the r.v.s $N_1(t_1), \ldots, N_1(t_n)$ are independent of $N_2(t_1), \ldots, N_2(t_n)$.

Combining Independent Poisson Processes

Definition 4

Two Poisson processes $\{N_1(t) : t > 0\}$ and $\{N_2(t) : t > 0\}$ are independent if for all $t_1, \ldots, t_n$, the r.v.s $N_1(t_1), \ldots, N_1(t_n)$ are independent of $N_2(t_1), \ldots, N_2(t_n)$.

Theorem 1

If $\{N_1(t) : t > 0\}$ and $\{N_2(t) : t > 0\}$ are independent Poisson processes of rates λ_1 and λ_2 and $N(t) = N_1(t) + N_2(t)$ for all $t > 0$, then $\{N(t) : t > 0\}$ is a Poisson process of rate $\lambda = \lambda_1 + \lambda_2$.

Combining Independent Poisson Processes

Idea of proof (using Definition 3): In any increment $(t, t+\delta]$,

$$\widetilde{N}(t, t+\delta) = \widetilde{N}_1(t, t+\delta) + \widetilde{N}_2(t, t+\delta)$$

has the stationary and independent increment property properties.

$$\begin{aligned} P_{\widetilde{N}(t,t+\delta)}(0) =& P_{\widetilde{N}_1(t,t+\delta)}(0) P_{\widetilde{N}_2(t,t+\delta)}(0) \\ =& [1-\delta\lambda_1 + o(\delta)][1-\delta\lambda_2 - o(\delta)] \\ =& 1-\delta(\lambda_1+\lambda_2) + o(\delta). \end{aligned}$$

$$\begin{aligned} P_{\widetilde{N}(t,t+\delta)}(1) =& P_{\widetilde{N}_1(t,t+\delta)}(1) P_{\widetilde{N}_2(t,t+\delta)}(0) + P_{\widetilde{N}_1(t,t+\delta)}(0) P_{\widetilde{N}_2(t,t+\delta)}(1) \\ =& [\delta\lambda_1 + o(\delta)][1-\delta\lambda_2 - o(\delta)] + [1-\delta\lambda_1 + o(\delta)][\delta\lambda_2 + o(\delta)] \\ =& \delta(\lambda_1+\lambda_2) + o(\delta). \end{aligned}$$

Combining Independent Poisson Processes

Idea of proof (using Definition 2): $N(t) = N_1(t) + N_2(t)$ is a Poisson rv., since for any given t, the sum of two independent Poisson rv.s is also a Poisson rv.. It can be derived by discrete convolution of the two PMF's.

$$P_{N(t)}(n) = \sum_{i=0}^{n} P_{N_1(t)}(i) P_{N_2(t)}(n-i) = \sum_{i=0}^{n} \frac{\lambda_1^i \lambda_2^{n-i}}{i!(n-i)!} t^n \exp(-\lambda_1 t - \lambda_2 t)$$
$$= \sum_{i=0}^{n} \binom{n}{i} \frac{\lambda_1^i \lambda_2^{n-i}}{n!} t^n \exp(-(\lambda_1 + \lambda_2)t) = \frac{(\lambda t)^n \exp(-\lambda t)}{n!}.$$

More elegantly, one can observe that we have already implicitly shown this fact. That is, if we break an interval I into disjoint subintervals, I_1 and I_2, then the number of arrivals in I (which is Poisson) is the sum of the number of arrivals in I_1 and I_2 (which are independent Poisson).

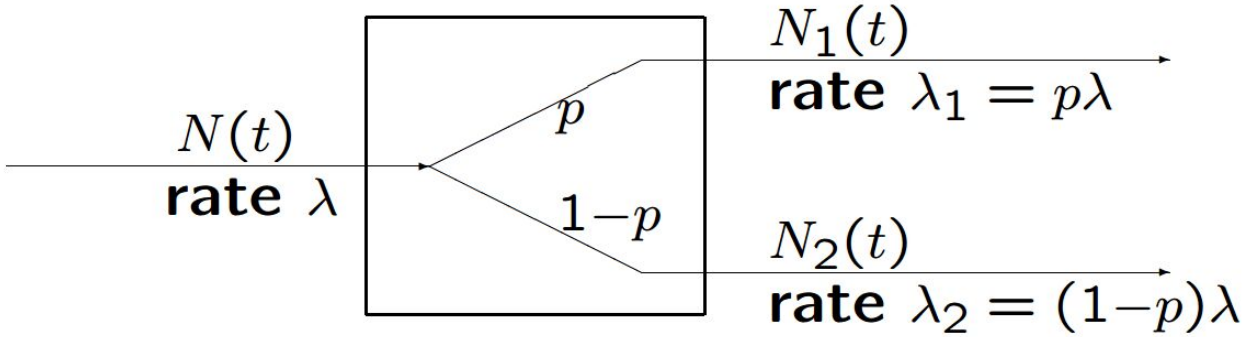
Combining Independent Poisson Processes

Idea of proof (using Definition 1): The first interarrival interval for the sum process, is the minimum of X_{11} and X_{21}, the first interarrival interval for the two independent processes.

$$P\{X_1 > t\} = P\{X_{11} > t\}P\{X_{21} > t\} = \exp(-\lambda_1 t - \lambda_2 t) = \exp(-\lambda t).$$

Remark. when many independent counting processes (even not Poisson) are added together, the sum process often tends to be approximately Poisson if the individual processes have small rates compared to the sum.

Subdividing a Poisson Process



Each arrival is switched to $\{N_1(t) : t > 0\}$ with probability p and otherwise goes to $\{N_2(t) : t > 0\}$. View the switch as a Bernoulli process independent of $\{N(t) : t > 0\}$. A p biased coin is flipped independently at each arrival.

We shall show that the resulting processes are independent Poisson rv.s with rates $\lambda_1 = \lambda p$ and $\lambda_1 = \lambda(1-p)$ respectively.

Subdividing a Poisson Process

- Each new process has the stationary and independent increment property and each satisfies the small increment property. Thus each is Poisson.
- To demonstrate the independence of these two processes,
  - calculate the joint PMF for $N_1(t)$, $N_2(t)$ for arbitrary t conditioning on a given number of arrivals $N(t)$ for the original process

$$P\{N_1(t) = m, N_2(t) = k | N(t) = m + k\} = \frac{(m+k)!}{m!k!} p^m (1-p)^k.$$

  - calculate the joint PMF for $N_1(t)$, $N_2(t)$ for arbitrary t

$$\begin{aligned} &P\{N_1(t) = m, N_2(t) = k\} = P\{N_1(t) = m, N_2(t) = k, N(t) = m + k\} \\ =&P\{N_1(t) = m, N_2(t) = k | N(t) = m + k\} P\{N(t) = m + k\} \\ =&\frac{(m+k)!}{m!k!} p^m (1-p)^k \frac{(\lambda t)^{m+k} \mathrm{e}^{-\lambda t}}{(m+k)!} = P\{N_1(t) = m\} P\{N_2(t) = k\}. \end{aligned}$$

This shows that $N_1(t)$ and $N_2(t)$ are independent.

Subdividing a Poisson Process

- To show that the processes are independent, we must show that for any $k > 1$ and any set of times $0 \leqslant t_1 \leqslant \cdots \leqslant t_k$, the sets $\{N_1(t_i) : 1 \leqslant i \leqslant k\}$ and $\{N_2(t_j) : 1 \leqslant j \leqslant k\}$ are independent of each other.
  It is equivalent to show that the sets $\{\widetilde{N}_1(t_{i-1}, t_i) : 1 \leqslant i \leqslant k\}$ and $\{\widetilde{N}_2(t_{j-1}, t_j) : 1 \leqslant j \leqslant k\}$ (where t_0 is 0) are independent.
  - The argument that $N_1(t)$ and $N_2(t)$ are independent shows this independence for $i = j$.
  - for $i \neq j$ the independence follows from the independent increment property of $\{N(t) : t > 0\}$.

Remark: any two independent Poisson processes can be viewed as being generated from a single process in this way.

Examples using independent Poisson processes

Example 1

Consider a last-come first-serve queue with Poisson arrivals, rate λ and independent exponential services, rate μ. A new arrival starts service immediately, but is interrupted if a new arrival occurs before service completion.

View services as a Poisson process. We can either ignore this process when there is nothing to serve, or visualize a low priority task that is served at rate μ when there is nothing else to do. The arrival process plus the service process is Poisson, rate $\lambda + \mu$.

Examples using independent Poisson processes

Example 1

Consider a last-come first-serve queue with Poisson arrivals, rate λ and independent exponential services, rate μ. A new arrival starts service immediately, but is interrupted if a new arrival occurs before service completion.

The probability an arrival completes service before being interrupted is $\mu/(\lambda+\mu)$.

Given that you are interrupted, what is the probability of no further interruption? Two services (the interrupting job and you) must finish before the next interruption, so Answer: $\frac{\mu^2}{(\lambda+\mu)^2}$.

Examples using independent Poisson processes

Example 2

Given independent Poisson processes $N_1(t)$ and $N_2(t)$ with the rate λ and μ relatively, find the probabilities $\Pr\{S_{1k} < S_{2j}\}$ where S_{1k} is the epoch of the kth arrival to the first process and S_{2j} is the epoch of the jth arrival to the second process.

The problem can be rephrased in terms of a combined process to ask: out of the first $k+j-1$ arrivals to the combined process, what is the probability that k or more of them are switched to the first process?

$$P\{S_{1k} < S_{2j}\} = \sum_{i=k}^{k+j-1} \binom{k+j-1}{i} p^i (1-p)^{k+j-1-i}.$$

Examples using independent Poisson processes

Example 3 (the M/M/1 queue)

Consider an M/M/1 queue, i.e., a queueing system with interarrival times are exponential with parameter λ and service times are exponential with parameter μ.

We see that if j or more customers are waiting at a given time, then Example 2 gives the probability that the kth subsequent arrival comes before the jth departure.

Poission processes

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Non-homogeneous Poisson Processes

A non-homogeneous Poisson process with time-varying arrival rate $\lambda(t)$ is defined as a counting process $\{N(t) : t > 0\}$ which has the independent increment property and, for all $t \geqslant 0$, $\delta > 0$, also satisfies:

$$P\{\widetilde{N}(t, t+\delta) = n\} = \begin{cases} 1 - \lambda(t)\delta + o(\delta) & \text{for} \quad n = 0, \\ \lambda(t)\delta + o(\delta) & \text{for} \quad n = 1, \\ o(\delta) & \text{for} \quad n \geqslant 2, \end{cases}$$

where $\widetilde{N}(t, t+\delta) = N(t+\delta) - N(t)$.

The non-homogeneous Poisson process does NOT have the stationary increment property.

Chapter 2. Poission processes

Review

§ Combining and splitting Poisson processes

§ Non-homogeneous Poisson processes

§ **Conditional arrival densities and order statistics**

Conditional arrival densities and order statistics

There are many interesting and useful results about the increment $(0, t]$ of a Poisson process conditional on $N(t)$. First condition on $N(t) = 1$.

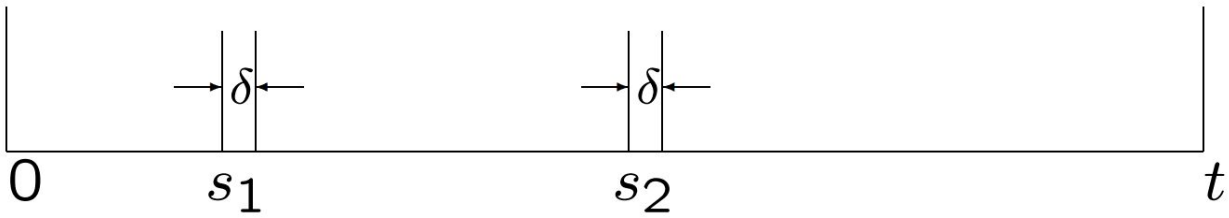
$$f_{S_1|N(t)}(s_1|1) = \lim_{\delta\to 0} \frac{P_{N(s_1)}(0)P_{\widetilde{N}(s_1,s_1+\delta)}(1)P_{\widetilde{N}(s_1+\delta,t)}(0)}{\delta P_{N(t)}(1)}$$

$$= \lim_{\delta\to 0} \frac{[\mathrm{e}^{-\lambda s_1}][\lambda\delta\mathrm{e}^{-\lambda\delta}][\mathrm{e}^{-\lambda(t-s_1-\delta)}]}{\delta\lambda t\mathrm{e}^{-\lambda t}} = \frac{1}{t}.$$

It does NOT depend on s_1 or λ. It is uniform over $(0, t]$.

Conditional arrival densities and order statistics

Case $N(t) = 2$.



$$f_{S_1,S_2|N(t)}(s_1, s_2|2) = \lim_{\delta\to 0} \frac{[\mathrm{e}^{-\lambda s_1}][\lambda\delta\mathrm{e}^{-\lambda\delta}][\mathrm{e}^{-\lambda(s_2-s_1-\delta)}][\lambda\delta\mathrm{e}^{-\lambda\delta}][\mathrm{e}^{-\lambda(t-s_2-\delta)}]}{\delta^2 P_{N(t)}(2)}$$
$$= \frac{2}{t^2}.$$

Again, this does not depend on s_1, s_2, or λ for $0 < s_1 < s_2 < t$, i.e., it is uniform over the given region of s_1, s_2.

Conditional arrival densities and order statistics

Theorem 2

Let $f_{S^{(n)}|N(t)}(s^{(n)}|n)$ be the joint density of $S^{(n)}$ conditional on $N(t) = n$. This density is constant over the region $0 < s_1 < \cdots < s_n < t$ and has the value

$$f_{S^{(n)}|N(t)}(s^{(n)}|n) = \frac{n!}{t^n}.$$

This is 'uniform' over $0 < s_1 < \cdots < s_n < t$.

It is a uniform n dimensional probability density over the volume $t^n/n!$ corresponding to the constraint region $0 < s_1 < \cdots < s_n < t$.

Conditional arrival densities and order statistics

Let $U_1, \ldots, U_n$ be n i.i.d. r.v.s, each uniform over $(0, t]$. Let $S_1, \ldots, S_n$ be the order statistics for $U^{(n)}$, i.e.,

$$S_1 = \min\{U_1, \ldots, U_n\}, \ldots, S_k = k\text{-th smallest}, \ldots$$

The region of volume t^n where the density of $U^{(n)}$ is nonzero partitions into $n!$ regions, one in which $u_1 < u_2 < \cdots < u_n$ and one for each other ordering of $u_1, \ldots, u_n$. From symmetry, each volume is the same, and thus each is $t^n/n!$.

Conditional arrival densities and order statistics

Another example of using order statistics: consider finding the distribution function of S_1 conditional on $N(t) = n$.

$$\begin{aligned} F^c_{S_1|N(t)}(s_1|n) &= \Pr\{\min_{1\leqslant i\leqslant n} U_i > s_1\} \\ &= \prod_{i=1}^{n} \Pr\{U_i > s_1\} \\ &= \prod_{i=1}^{n} \left(1 - \frac{s_1}{t}\right) = \left(1 - \frac{s_1}{t}\right)^n \quad \text{for } s_1 \leqslant t. \end{aligned}$$

Thus, $\mathrm{E}[S_1|N(t) = n]$ can be found by integration,

$$\begin{aligned} \mathrm{E}[S_1|N(t) = n] &= \int_0^t F^c_{S_1|N(t)}(\tau|n)\mathrm{d}\tau \\ &= \frac{t}{n+1}. \end{aligned}$$

Conditional arrival densities and order statistics

For S_2 to S_n, it is slightly simpler to find $f_{S_i|N(t)}(s_i|n)$.

Look at n uniformly distributed r.v.s in $(0, t]$

- The probability that one of these lies in the interval $(s_i, s_i + \mathrm{d}t]$ is $\frac{n}{t}\mathrm{d}t$.
- Out of the remaining $n-1$, the probability that $i-1$ lie in the interval $(0, s_i]$ is given by the binomial distribution with probability of success s_i/t. Thus the desired density is

$$f_{S_i|N(t)}(s_i|n)\mathrm{d}t = \binom{n-1}{i-1}\left(\frac{s_i}{t}\right)^{i-1}\left(1-\frac{s_i}{t}\right)^{n-i}\frac{n}{t}\mathrm{d}t,$$

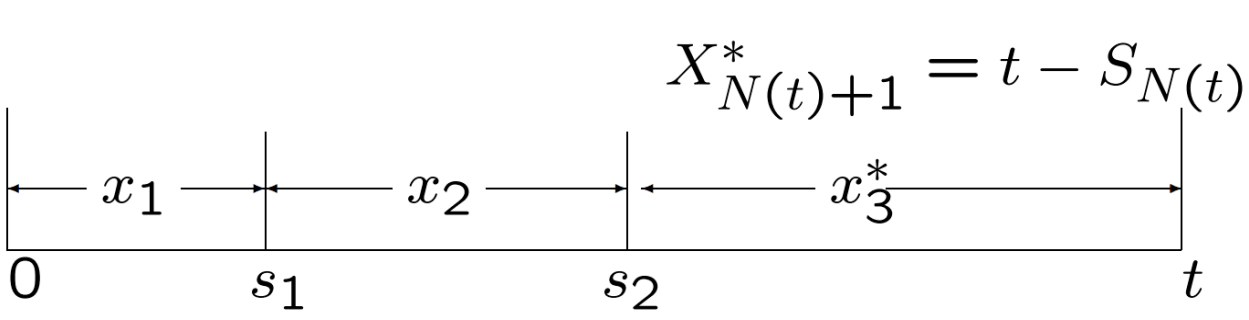
$$f_{S_i|N(t)}(s_i|n) = \frac{s_i^{i-1}(t-s_i)^{n-i}n!}{t^n(n-i)!(i-1)!} \quad \text{for } s_i \leqslant t.$$

Conditional arrival densities and order statistics

Return to $N(t) = 2$ and look at $f_{X_1,X_2|N(t)}(x_1, x_2|2)$. Note that the area in the s_1, s_2 space where $0 < s_1 < s_2 < t$ is $t^2/2$, explaining why the density is $2/t^2$.

Note that a δ^2 box in the s_1, s_2 space maps into a parallelepided of the same area in the x_1, x_2 space. The area in the x_1, x_2 space where $0 < x_1 + x_2 < t$ is again $t^2/2$ and $f_{X_1,X_2|N(t)}(x_1, x_2|2) = 2/t^2$.

Order Statistics



$$f_{X_1,X_2|N(t)}(x_1,x_2|2) = \frac{2}{t^2} \quad \text{for } x_1, x_2, x_3^* > 0;\ x_1 + x_2 + x_3^* = t.$$

From symmetry, any 2 of the variables X_1, X_2, X_3^* can replace X_1, X_2 above.

Also, since $X_1 = S_1$, X_1 has density and expected value

$$F^c_{X_1|N(t)}(x_1|2) = \left(1 - \frac{x_1}{t}\right)^2, \qquad \mathrm{E}[X_1|N(t) = 2] = \frac{t}{3}.$$

We see that X_2 and X_3^* can be substituted for X_1 in the above formulas.

Conditional arrival densities and order statistics

Extend to the $N(t) = n$ condition, we have

$$f_{X^{(n)}|N(t)}(x^{(n)}|n) = \frac{n!}{t^n} \quad \text{for } X^{(n)} > 0, \ \sum_{i=1}^{n} X_i < t.$$

For $i = 1, 2, \ldots, n$,

$$F^c_{X_i|N(t)}(x|n) = \left(1 - \frac{x}{t}\right)^n.$$

$$\mathrm{E}[X_i|N(t) = n] = \frac{t}{n+1}.$$

Therefore,

$$\mathrm{E}[S_i|N(t) = n] = \frac{it}{n+1}.$$

Order Statistics

This relation also applies to $X^*_{n+1} = t - S_{N(t)}$

$$F^c_{X^*_{n+1}|N(t)}(x|n) = \left(1 - \frac{x}{t}\right)^n.$$

Thus,

$$\mathrm{E}[X^*_{n+1}|N(t) = n] = \frac{t}{n+1}.$$

$$\begin{aligned}\mathrm{E}[X^*_{N(t)+1}] &= \sum_{n=0}^{\infty} \frac{t}{n+1} \frac{(\lambda t)^n \exp(-\lambda t)}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(\lambda t)^{n+1} \exp(-\lambda t)}{\lambda (n+1)!} = \frac{1 - \exp(-\lambda t)}{\lambda}.\end{aligned}$$

The mean time back to the previous arrival is $\frac{1-\exp(-\lambda t)}{\lambda}$.

Conditional arrival densities and order statistics

Recall that the joint density of interarrivals $X_1, \ldots, X_n$ and arrival epoch S_{n+1} is

$$f_{X_1 \cdots X_n S_{n+1}}(x_1, \ldots, x_n, s_{n+1}) = \lambda^{n+1} \exp(-\lambda s_{n+1}).$$

Conditional on $S_{n+1} = t$ (which is Erlang),

$$f_{X_1 \cdots X_n | S_{n+1}}(x_1, \ldots, x_n | t) = \frac{\lambda^{n+1} \exp(-\lambda t)}{\lambda^{n+1} t^n \exp(-\lambda t)/n!} = \frac{n!}{t^n}.$$

On the other hand,

$$f_{X_1 \cdots X_n | N(t)}(x_1, \ldots, x_n | n) = \frac{n!}{t^n}.$$

Both equations are for $0 < x_1, \ldots, x_n$ and $\sum_{i=1}^n x_i < t$.

Both say the conditional density is uniform over the constraint region.

Why are the two equations the same?

课堂练习

1. 假设某零件受到外来撞击的次数服从参数为 λ 的 Poisson 过程。已知该零件受到撞击次数超过 k 时就不能正常工作，求该零件的使用寿命 T 的分布密度函数。

课堂练习

2. 甲乙丙丁玩游戏。掷一对均匀的骰子，其点数和记为 X，设所掷次数服从平均每小时 6 次的泊松分布。一次游戏中 $X \mod 4 = 1, 2, 3, 0$ 时，分别对应甲、乙、丙、丁胜。

(1) 计算两小时内，甲胜的次数；

(2) 计算甲、丙在两个小时内共胜的次数；

(3) 若两小时内做了 12 次的游戏，甲胜次数；

(4) 若第一小时内甲胜了三次，问两小时内共玩了多少次的游戏。

课堂练习

3. 设某网店经营 A,B 两种商品。A 商品下单到达是单位时间 4 单，B 商品下单到达是单位时间 2 单，均是泊松过程。求：

(1) 前三个单位时间内收到 3 单的概率；

(2) 已知前三个单位时间内收到 3 单，求至少有一个是 B 订单的概率；

(3) 第 2 个订单是 B 订单的概率；

(4) 第 2 个订单到达时间在第二个单位时间内的概率。

课堂练习

4. 甲乙两人在做投硬币游戏。甲以速率为 $\lambda_1 = 2$ 的泊松过程投一枚不均匀硬币，其正面向上概率为 3/4。乙以速率为 $\lambda_2 = 3$ 的泊松过程投一枚不均匀硬币，其正面向上概率为 2/3。两人投硬币的过程相互独立。求

(1) 甲比乙先投出 1 次正面向上的概率；

(2) 甲第 1 次投出正面向上时，乙已经投出正面向上次数的分布；

(3) 若甲乙两人共投出 5 次正面向上时，游戏结束。求游戏持续时间的期望，以及游戏结束时，两人共投硬币次数的期望。

课堂练习

5. 设 U_1, U_2, U_3, U_4, U_5 是 5 个在 $[0,1]$ 上独立均匀分布的随机变量，X_1, X_2, X_3, X_4, X_5 是 $\{U_i, 1 \le i \le 5\}$ 的顺序统计量，即 X_i 为 $\{U_i, 1 \le i \le 5\}$ 从小到大排序的第 i 个值，

$$X_1 = \min\{U_1, U_2, U_3, U_4, U_5\}, \ldots, X_5 = \max\{U_1, U_2, U_3, U_4, U_5\}.$$

计算

(1) $\mathrm{E}[X_5]$;

(2) $\mathrm{E}[X_4 | X_5 = 0.8]$;

(3) $\mathrm{E}[X_4 | X_5 = x_5], x_5 \in [0,1]$;

(4) $\mathrm{E}[X_4]$;

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6. 令 $\mathrm{N} = (N(t), t \geq 0)$ 是参数为 λ 的 Poisson 过程，$S_1, S_2, \ldots$ 表示顾客达到的时刻，求

(1) $E(S_1 S_2 | N(1) = 2)$;

(2) $E(S_1 + S_2 + \cdots + S_5 | N(1) = 5)$。

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7. 令 $\mathrm{N} = (N(t), t \geq 0)$ 是参数为 λ 的 Poisson 过程，$S_1, S_2, \ldots$ 表示顾客达到的时刻，令

$$U = \frac{S_1}{S_2}, V = \frac{1 - S_3}{1 - S_2}$$

已知 $N(1) = 3$，求 (U, V) 的联合分布。

Thank you for your attention!